CIS 5190 Applied Machine Learning		Spring 2026 News Source Project

Beyond Words:
Stylistic Signals in News Headline Source Classification

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1 Summary

In this project, we investigate whether machine learning models can accurately classify digital news headlines from NBC News and Fox News, two of the most prominent outlets in American news media. Because these organizations cover the news from opposing editorial perspectives, we might expect these differences to show up in their headlines. But beyond vocabulary and topical content, we further hypothesized that outlet signals would be present in the stylistic conventions each organization follows: punctuation choices, capitalization standards, and formatting patterns. This paper documents our approach to building a binary classifier that predicts a headline's source using only its text, starting with a simple TF-IDF baseline and advancing to a stacking ensemble that achieves 81.2% accuracy on a held-out test set. Our results show that stylistic signals provide meaningful predictive power to complement traditional vocabulary-based features.

2 Data Collection and Dataset

Scraping. We assembled our headline corpus in two phases. We began with a course-supplied list of 3,805 article URLs split between Fox News and NBC News, scraping each with `requests` and `BeautifulSoup` and extracting the article headline from `<h1 class="headline speakable">` on Fox pages and the leading `<h1>` on NBC pages. URLs returning a non-200 status or lacking a recognizable headline were marked missing and dropped at the cleaning stage, yielding our base scrape.

Expansion. After our first leaderboard submission revealed a substantial accuracy drop relative to local cross-validation, we hypothesized distribution shift: the hidden test set appeared to draw from a later news cycle than the course-supplied URL list, leaving our base scrape stale relative to the headlines it was being graded against. To narrow that gap, we expanded the corpus by querying Google News RSS one `(source, topic, day)` slice at a time, combining the `site:foxnews.com` and `site:nbcnews.com` operators with twenty broad topics (politics, immigration, economy, foreign policy, sports, and similar) across every calendar day from 2025-11-01 through 2026-04-30; per-day slicing keeps each query under Google's roughly 100-result cap. To prevent label leakage, we stripped trailing source-identifying suffixes such as "- Fox News and "—NBC News" from RSS titles before retaining them. We then deduplicated the new rows exactly within source, downsampled per source to balance the new Fox/NBC counts, and removed any near-duplicates of the base scrape using a normalized (lowercased, alphanumerics-only) key. Concatenating the survivors with the base rows produced the expanded dataset that all downstream modeling uses.

Cleaning. Our data cleaning was guided by exploratory data analysis of the raw scraped text, which revealed several categories of invalid entries. We first addressed Spanish-language content scraped from Fox's Latino News pages by implementing a two-step filter combining Spanish word/character frequency counts with a ratio-based threshold. We then located and removed structural artifacts, including index pages, radio schedules, and podcast and episode titles. As a final safeguard, we dropped headlines outside a 25–140-character range, as we found these were typically bylines, page titles, or episode descriptions that would only introduce noise to our models (see Appendix Figures A1 and A2). After cleaning, our dataset contained approximately 26,900 headlines (Fox: 12,338; NBC: 14,553).[1]

To address the moderate class imbalance in our cleaned dataset, all of our modeling used a stratified 80/20 train/test split.

[1] Submitted as the Datasets exploratory component.

3 Exploratory Data Analysis

Before modeling, we investigated whether Fox and NBC headlines differed systematically in style.[2] This hypothesis was motivated by one group member's background in journalistic copy editing, where an organization's style guide governs punctuation, capitalization, and abbreviation standards, and may even set expectations for headline length and structure — suggesting these signals might vary predictably across outlets.

To test this, we computed chi-squared statistics and Cramer's V effect sizes across a set of candidate formatting features extracted from our raw headline text. While we had already removed non-English content at this stage, we intentionally conducted this analysis before applying our full cleaning pipeline, since that would have removed many of the stylistic cues we were interested in. Our findings confirmed our hypothesis: Several style features showed statistically significant differences between sources, as shown in Table 1. On average, Fox headlines tended to be longer, used more colons and capital letters (including all-caps words), and favored "US" when referencing the United States. NBC headlines by contrast were more likely to use periods, both as a general punctuation style (including headline-ending periods) and in following the Associated Press Stylebook convention of "U.S." with internal punctuation.[3]

Table 1: Headline Style Feature Significance Testing (Chi-squared and Cramer's V)

Feature	χ^2	p	**Cramer's V**	**NBC mean**	**Fox mean**
Continuous					
Length	3171.66	<0.001	0.333	72.0	**78.4**
Capital letter count	1787.39	<0.001	0.250	3.8	**5.3**
Period count	1448.73	<0.001	0.225	**0.3**	0.1
All-caps word count	400.64	<0.001	0.118	0.3	**0.4**
Binary					
U.S.	882.33	<0.001	0.176	**7.3%**	0.4%
Is title case	817.70	<0.001	0.169	0.7%	**7.3%**
US	721.23	<0.001	0.159	0.2%	**5.3%**
Contains colon	526.81	<0.001	0.136	11.5%	**21.5%**
Ends with period	297.20	<0.001	0.102	**2.3%**	0.1%
Contains double quote	187.15	<0.001	0.081	27.1%	**34.5%**
Contains single quote	170.21	<0.001	0.077	27.0%	**34.1%**
Contains hyphen	47.94	<0.001	0.041	9.6%	**12.1%**

These findings drove our modeling approach. Rather than stripping all punctuation and formatting during preprocessing, we designed our pipeline to retain these signals wherever possible. We normalized text by converting to lowercase and resolving encoding inconsistencies but otherwise deliberately preserved the stylistic differences we identified as predictive. This analysis ultimately motivated two key components of our pipeline: a character n-gram branch to capture subword punctuation patterns, and a dedicated style feature branch operating on raw text to encode the 12 handcrafted features summarized in Table 1.

4 Model Design

We developed our model in two phases so that gains could be attributed to specific changes: first, holding the classifier fixed at logistic regression and iterating on the feature representation, then holding the best feature pipeline fixed and iterating on the classifier.

Feature Engineering. Working from a stopword-filtered TF-IDF baseline of 100 features (macro F1 = 0.5805), we found that applying our text normalizer alone yielded no improvement (F1 = 0.5792), indicating

[2] Submitted as the Analysis exploratory component.
[3] These signals reflect editorial practices at the time of data collection; we note that style guides evolve, and model performance may degrade if outlets update their conventions over time.

that capacity, not normalization, was the binding constraint. Scaling the vocabulary to 5,000 features with bigrams and sublinear term frequencies jumped performance to F1 = 0.7389; layering on character `wb` n-grams of length 3–5 captured the subword punctuation and capitalization signals identified in our EDA, lifting F1 to 0.7583. Combining the word and character branches via `FeatureUnion` (our `Hybrid` pipeline) reached F1 = 0.7766, and adding our 12-feature style branch on *raw* text, since style features rely on caps and punctuation that the normalizer would otherwise smooth away, pushed our `Hybrid_v2_style` pipeline to F1 = 0.7899. A feature-level grid search on this final pipeline confirmed 5,000 word features, word n-grams of $(1, 2)$, and character n-grams of $(4, 6)$ as optimal, giving F1 = 0.7939.

Table 2: Feature-Pipeline Progression (5-fold CV on training split)

Pipeline	Description	Macro F1	Accuracy
Baseline	TF-IDF, 100 features, English stopwords, LR	0.5805	0.6101
Cleaned	Baseline + text normalization	0.5792	0.6100
Optimized_v2	5,000 features, $(1, 2)$ word n-grams, sublinear TF	0.7389	0.7435
Optimized_v2_tokenized	Optimized_v2 + normalization + custom token pattern	0.7407	0.7450
V3_char_grams	TF-IDF on $(3, 5)$ `char_wb` n-grams	0.7583	0.7611
Hybrid	Word + character branches (`FeatureUnion`)	0.7766	0.7791
Hybrid_v1	Hybrid with widened word vocab and $(4, 6)$ chars	0.7751	0.7777
Hybrid_v2_style	Word + char + 12 style features on raw text	0.7899	0.7919
Hybrid_v2_style (tuned)	Same, after feature-level grid search	**0.7939**	**0.7966**[4]

Classifier Selection and Hyperparameter Tuning. With the feature pipeline fixed, we swept nine classifier families. Logistic regression led at F1 = 0.7939, with SGD (0.7875), LinearSVC (0.7707), and random forest (0.7677) close behind; the Naive Bayes variants landed near 0.73, while AdaBoost, decision trees, and KNN trailed in the 0.65–0.69 range. That the linear models clustered at the top suggested that representation, not classifier capacity, was the binding constraint, consistent with what we had observed in phase 1. A hyperparameter grid search over the top four classifiers produced modest but uneven gains: LinearSVC saw the largest absolute lift (+0.022 to F1 = 0.7929), followed by SGD (+0.010 to 0.7972), random forest (+0.006 to 0.7737), and logistic regression, which was essentially flat at its default optimum.[5]

Ensemble Methods. Finally, we explored three ensembling strategies on top of the four tuned base learners. A soft vote over the three probability-native classifiers (LR, SGD, RF) reached F1 = 0.8024; adding LinearSVC via `CalibratedClassifierCV` gave F1 = 0.8013; and a stacking classifier feeding all four into a logistic-regression meta-learner topped the group at F1 = 0.8030 (Figure 1). Stacking won by only a hair, but we chose it because the meta-learner can learn non-uniform weights over base classifiers rather than averaging them, which is a better fit for a setting where the base models differ noticeably in CV F1.The final submission therefore combines the `Hybrid_v2_style` feature pipeline with the four-learner stacking ensemble.

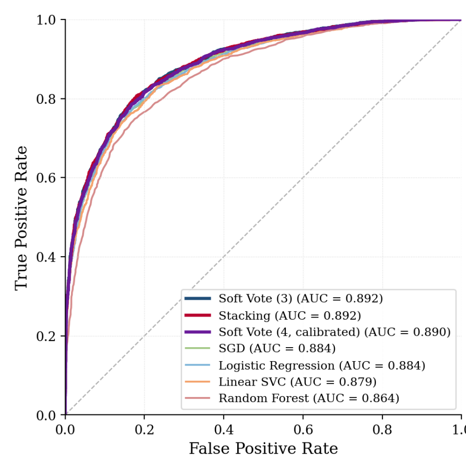

Figure 1: ROC curves for the four tuned classifiers and three ensembles on the held-out test split.

[4] CV accuracy estimated separately via `cross_val_score`.

[5] We excluded XGBoost from this sweep because the grading environment ships only `numpy`, `pandas`, `torch`, `scikit-learn`, and `opencv-python`, so a pipeline including `xgboost` would fail at load.

Neural Exploration. To investigate whether learned representations could improve on our traditional ensemble, we built a lightweight neural classifier in PyTorch.[6] Unlike TF-IDF, which treats tokens independently, this model captures richer token-level features by mapping inputs to a 128-dimensional embedding, followed by a pooling layer and a two-layer MLP. We evaluated a text-only model and a hybrid model that concatenates our 12 stylistic features prior to the classification layer. The hybrid model achieved a macro F1 of 0.781 on the held-out test set, outperforming the text-only model (F1 = 0.769) and confirming that stylistic signals provide gains across model families. However, both models underperformed our stacking ensemble. Given the shortness of headlines and our dataset's scale, this was not particularly surprising: Our well-tuned traditional models are more competitive than neural architectures under these limitations.

5 Evaluation and Model Performance

We promoted models at each phase boundary by 5-fold cross-validation macro F1 on the training split, using macro F1 as our primary metric throughout. Our neural model was evaluated solely on the held-out test set to avoid the computation required for cross-validation over the full training loop. The final stacking ensemble was evaluated once on the held-out 20% test split. On the test split (n = 5,379), our submission reached **81.2% accuracy** with a macro F1 of 0.81. Per-class scores were balanced: NBC at precision 0.82, recall 0.84, F1 0.83 (n = 2,911); Fox at precision 0.81, recall 0.78, F1 0.79 (n = 2,468) — the model is slightly more conservative on Fox, given that precision exceeds recall. Test accuracy (0.8119) tracks the CV macro F1 of the same pipeline (0.8030) closely, suggesting our held-out estimate is well-calibrated and that we are not overfitting through grid search or ensemble selection.

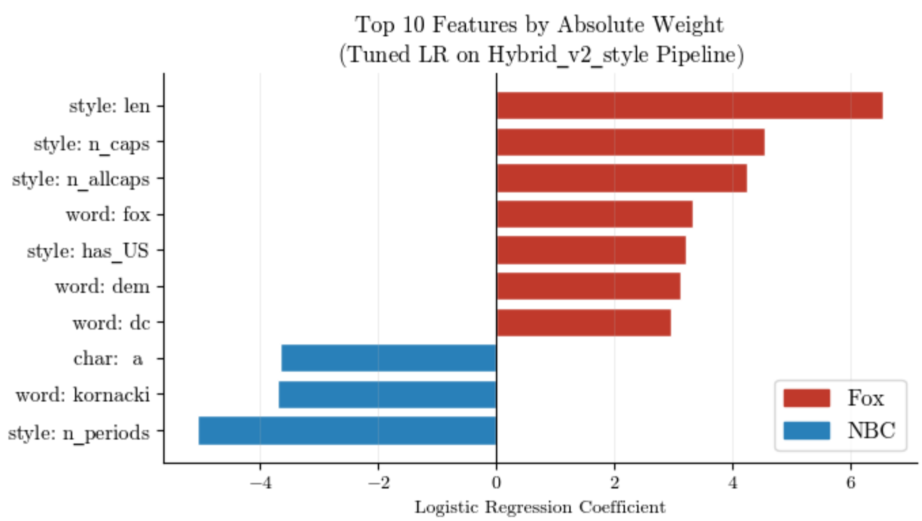

Figure 2: Top 10 features by absolute weight for the tuned LR pipeline.

Inspecting the logistic regression coefficients of the tuned `Hybrid_v2_style` pipeline confirms our central EDA hypothesis: Stylistic features dominate the highest-weighted coefficients (Figure 2). Five of the ten highest-weighted features are handcrafted style signals, including headline length (+6.54), period count (-5.05), capitalization counts, and the binary "US" indicator (+3.20), recovering the same patterns surfaced by chi-squared testing in Table 1. The remaining top features are vocabulary-based, with `fox` and `dem` weighted toward Fox and `kornacki` weighted toward NBC.[7]

[6] Submitted as the Technique exploratory component.

[7] Named entities like `kornacki` illustrate a distribution shift concern: Predictive vocabulary in our 2025-11 to 2026-04 window may degrade as organization personnel and news cycles change over time.

6 Conclusion

Our study demonstrates that news headlines carry distinct, learnable stylistic signatures that enable reliable source classification from text alone. Our stacking ensemble achieved 81.2% accuracy and 0.81 macro F1 on the held-out test set, with the bulk of that improvement coming from feature engineering rather than classifier selection. The improvement from a simple TF-IDF baseline (macro F1 = 0.58) to our final model (+0.23 macro F1) was driven overwhelmingly by scaling vocabulary capacity, adding character n-grams, and incorporating handcrafted style features. Classifier selection and hyperparameter tuning contributed modestly by comparison.

The central finding of our work is that stylistic conventions are genuinely predictive signals. Beyond topical differences, we show that Fox and NBC headlines diverge systematically in capitalization, punctuation, and formatting that align with distinct editorial standards. We identified these signals during EDA, designed our cleaning pipeline to preserve them, and confirmed they dominated the learned representation through coefficient analysis of our final ensemble's underlying logistic regression. The consistency of these signals across both traditional and neural models reinforces this finding.

Several limitations should be noted. Our dataset captures a single time window, and the model partially depends on time-sensitive vocabulary (e.g., anchor names, ongoing conflicts, and current political figures) that may not generalize to future news cycles. Similarly, style conventions could evolve as outlets update their guidelines. Future work should examine the temporal stability of both stylistic and topical features using longitudinal or rolling datasets, or explore pretrained transformer models to better disentangle style from content at larger scales.

Team Contributions

Katie designed our data preprocessing workflows, conducted EDA and translated insights to the modeling and evaluation framework, and conducted the neural model exploration. Isaac drove our data collection strategy, optimized model performance through classifier sweeps and ensemble experiments, and oversaw all codebase integration and technical submission requirements. Both authors collaborated in preparing the final report.

Appendix

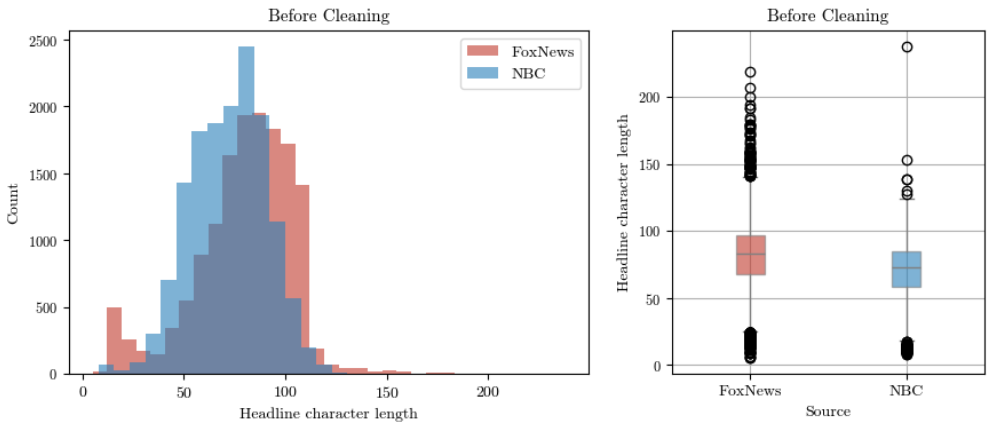

Figure A1: Headline length distribution by source before cleaning.
Note the cluster of very short Fox headlines (bylines, page titles) and the long right tail from Fox News Radio and other streaming/video episode titles.

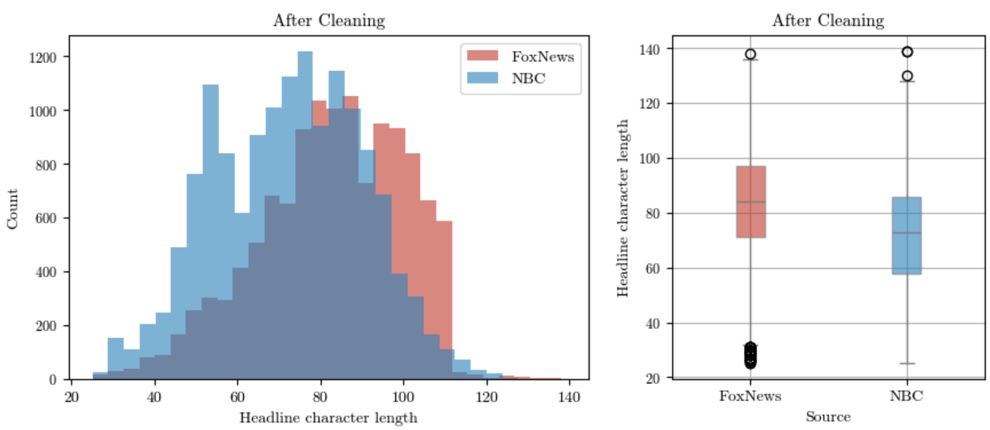

Figure A2: Headline length distribution by source after cleaning.
Both sources now follow approximately normal distributions within the 25–140 character range. Fox headlines remain slightly longer on average, consistent with the Cramer's V = 0.333 reported in Table 1.

Acknowledgements

We used AI assistance for general debugging, formatting code comments and section summaries, building plotting and caching helper functions, migrating our original analysis files into `model.py` and `preprocess.py`, and constructing the loop to combine Google News RSS queries with our baseline dataset.